Related-Work Table - BUSI-WHU (CSE475 Task 1)

Group Group08 | Members: Maharun Nesa Tanni (2023-1-60-190), Kh. Tania Akter Tamanna (2023-2-60-070) | Track 3 (CNN + Attention) | 5 papers, 2024-2025. All five papers have open-access PDFs (in related_work/papers).

#	Title	Authors	Year / Venue	Dataset	Application	Method / model	Metrics (as reported)	Strengths	Limitations	Research gap	Relation to our work	Track 3: attention type
1	Node-Based Graph Convolutional Network With SLIC Method for Breast Cancer Ultrasound Images Classification	Trang, Ting, Vuong, Ting	2024 / IEEE Access, vol. 12	BUSI, BUS-BRA	Breast US classification (benign / malignant / normal)	SLIC superpixels as nodes; edges by colour similarity + spatial proximity; multi-layer GCN; node voting for image label	Classification metrics on both datasets (see paper)	Real image-to-graph pipeline; studies number of connections between nodes	Superpixels cover whole image (background nodes); not tested on BUSI-WHU; patient-wise split not stated	No tumour-mask-guided graph; no leakage-aware evaluation	Graph-based classification reference (different family from our attention CNN)	None
2	Malignancy pattern analysis of breast ultrasound images using clinical features and a graph convolutional network	Montaha, Azam, Bhuiyan, Chowa, Mukta, Jonkman	2024 / Digital Health	Breast US image datasets (A, B, A+B)	Benign vs malignant classification	Graph of clinical markers (shape, orientation, margin, surrounding tissue); Pearson-correlation edges; GCN vs GAT	Accuracy 91.03% / 94.37% / 89.62% on supplementary datasets A / B / A+B	Compared with GAT, 1D-CNN, transfer learning, 10 ML and 3 ensemble models	Handcrafted markers, not image regions; nodes are features, not places in the image	Graph is not spatial; no segmentation-to-graph link; no CV significance test reported	Graph-based classification reference; shows tumour shape features are informative	None
3	HyFormer-Net: A Synergistic CNN-Transformer with Interpretable Multi-Scale Fusion for Breast Lesion Segmentation and Classification in Ultrasound Images	Rahman, M. A.	2025 / arXiv:2511.01013	BUSI (+ external set for transfer)	Segmentation + classification, XAI	EfficientNet-B3 + Swin dual-branch encoder, multi-scale fusion, attention-gated decoder	BUSI Dice 0.761 +/- 0.072, classification acc 93.2%; attention gates +5.9% Dice; zero-shot transfer Dice 0.058	Ablation of attention; Grad-CAM explainability; cross-dataset test	Preprint (not peer reviewed); no BUSI-WHU; poor cross-dataset transfer	Domain shift between datasets is not solved	Track 3 reference for ablation + Grad-CAM style analysis	Attention gates + Swin self-attention (hybrid)
4	Hybrid Attention Network for Accurate Breast Tumor Segmentation in Ultrasound Images	Aslam, Naveed, Ahmed	2025 / arXiv:2506.16592	BUSI (780 imgs), UDIAT (163 imgs)	Breast US tumour segmentation	DenseNet121 encoder + attention decoder (global spatial attention, position encoding, scaled dot-product attention, SFEB); BCE + Jaccard loss	Jaccard, Dice, sensitivity, precision, accuracy, specificity; BUSI Dice 97.28%, Jaccard 94.75% (as reported)	Ablation table; Friedman + Nemenyi significance test; low parameter count	Segmentation only; no data augmentation; fixed train/test split with no patient grouping mentioned; preprint	No classification; no explainability; not tested on BUSI-WHU	Track 3 reference: attention ablation and significance test design	Global spatial + self-attention + semantic attention (hybrid)
5	A dual-branch and dual attention transformer and CNN hybrid network for ultrasound image segmentation (DDTransUNet)	Zhang, Wang, Wei, Kong, Qiu	2024 / Frontiers in Physiology, vol. 15	Three US datasets incl. TN3K and CAMUS	Ultrasound organ / lesion segmentation	Swin Transformer + CNN dual-branch encoder; global spatial + channel attention; cross-attention fusion	IoU, Dice, HD95, ACC; TN3K Dice 82.31%, CAMUS Dice 90.33%	Spatial + channel attention and cross-attention fusion; open access	Not breast-specific; segmentation only; no BUSI-WHU	No classification or graph; breast benign/malignant not studied	Track 3 reference for spatial vs channel attention comparison	Spatial + channel attention + cross-attention (hybrid)

Also reviewed (abstract level, paywalled PDFs): DSATNet (Medical Physics 2025) and DFLNet (Digital Signal Processing 2025) both use BUSI-WHU for segmentation only. **Research gap (from these papers):** (1) The papers we found that use BUSI-WHU (DSATNet, DFLNet) do segmentation only; none benchmark benign-vs-malignant classification on BUSI-WHU. (2) Image-to-graph GCN work (Trang et al.) uses BUSI/BUS-BRA with whole-image SLIC superpixels, not tumour-mask-guided region graphs; Montaha et al. use feature-node graphs, not image regions. (3) Patient-grouped evaluation, 5-fold CV with a significance test, and explainability are rarely reported together (only Aslam et al. test significance, only HyFormer-Net uses Grad-CAM). **Our contribution:** a patient-grouped, leakage-free comparison of CNN baselines and our proposed CNN with an attention module (Track 3) on BUSI-WHU, with ablation, 5-fold CV, significance test and XAI.